Quiz 3: Practice

15 minutes 10 points

Name: _____ **UW NetID:** _____ **Campus ID:** _____

Final answers **For TA:** Total _____ / 10

1(a)	1(b)	2(a)	2(b)	2(c)

Lectures 8–11. Not graded. Closed book, with the formulas below. Work individually; no personal notes or AI tools. **Write final answers in the table above:** numbers for 1(a)–(b), one letter (A–D) for each of 2(a)–(c). Exact fractions or decimals are accepted. Each part is worth 2 points; no partial credit.

Reference. For the transition $(S_t, A_t, R_{t+1}, S_{t+1})$ and step size α:

$$V(S_t) \leftarrow V(S_t) + \alpha\big(\text{target} - V(S_t)\big), \qquad Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big(\text{target} - Q(S_t, A_t)\big);$$

targets: TD(0) $R_{t+1} + \gamma\, V(S_{t+1})$, SARSA $R_{t+1} + \gamma\, Q(S_{t+1}, A_{t+1})$, Q-learning $R_{t+1} + \gamma \max_a Q(S_{t+1}, a)$.

Monte Carlo return: $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$.
Certainty equivalence: $\widehat{P}, \widehat{r}$ by counts and averages; in the estimate, $\widehat{Q}^\pi(s,a) = \widehat{r}(s,a) + \gamma \sum_{s'} \widehat{P}(s' \mid s,a)\, \widehat{V}^\pi(s')$ and $\widehat{V}^\pi(s) = \widehat{Q}^\pi(s, \pi(s))$.

1. One update, two targets **4 points**

Let $\alpha = \gamma = \frac{1}{2}$, and suppose

$$Q(\mathrm{X}, a) = 2, \qquad Q(\mathrm{Y}, a) = 4, \qquad Q(\mathrm{Y}, b) = 8.$$

The only actions at Y are a and b. The agent takes a in X, receives reward 4, moves to Y, and chooses a next. Start each part from the values above.

(a) Q-learning. Apply one update. Give the new $Q(\mathrm{X}, a)$.

(b) SARSA. Apply one update. Give the new $Q(\mathrm{X}, a)$.

2. Estimates, targets, and policies **6 points**

(a) In a finite discounted MDP, SARSA and Q-learning both use the exact table $Q^\star$ to form their targets. Data come from a fixed behavior policy π_b, which need not be optimal; SARSA's next action is drawn from π_b. For which rule is the following equality guaranteed at every visited pair?

$$\mathbb{E}\big[\text{target} \;\big|\; S_t = s, A_t = a\big] = Q^\star(s,a).$$

A. Both **B.** SARSA only **C.** Q-learning only **D.** Neither

(b) TD(0) evaluates a fixed policy π on a continuing task. The current estimates satisfy $V(s_0) = V^\pi(s_0) + c$ and $V(s') = V^\pi(s')$ for every $s' \neq s_0$. Given $S_t = s$, what is the mean of the target $R_{t+1} + \gamma V(S_{t+1})$? ($P^\pi(s_0 \mid s)$: the chance that $S_{t+1} = s_0$.)

A. $V^\pi(s) + \gamma c$ **B.** $V^\pi(s) + \gamma P^\pi(s_0 \mid s)\, c$ **C.** $V^\pi(s)$ **D.** $V^\pi(s) + P^\pi(s_0 \mid s)\, c$

(c) Consider Q-learning in a finite MDP with deterministic rewards and transitions, $0 < \gamma < 1$, and fixed $\alpha \in (0, 1]$. The agent always acts greedily, with fixed tie-breaking. After some time, its table Q remains exactly unchanged, while every state is still visited infinitely often. Let π_g be the greedy policy of this unchanged table. Which statements must hold?

I. $Q(s, \pi_g(s)) = V^{\pi_g}(s)$ for every state s.
II. π_g is optimal.

A. Both I and II **B.** I only **C.** II only **D.** Neither